CUSTOMER PERSONALITY SEGMENTATION

Unsupervised Machine Learning for Customer Clustering & Behavioral Segmentation

Project Report

AI / Machine Learning — Feature Engineering & Clustering Module

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Executive Summary

This report documents an end-to-end unsupervised machine learning project focused on customer personality segmentation. The objective was to group customers into meaningful behavioral segments using clustering techniques, enabling data-driven marketing and customer relationship strategies.

The project followed a complete data science pipeline: data validation, feature selection, feature scaling, exploratory analysis, determination of optimal cluster count, model building across four clustering algorithms (K-Means, Hierarchical/Agglomerative Clustering, Gaussian Mixture Models, and DBSCAN), rigorous evaluation and comparison, cluster profiling, stability testing, and final model selection with production-ready artifact export.

K-Means clustering was selected as the final production model, producing two well-separated, statistically stable customer segments — labeled "High-Value Customers" and "Premium Buyers" — with a Silhouette Score of 0.2516 and a Davies-Bouldin Index of 1.5371.

1. Introduction & Objective

Customer segmentation is a core application of unsupervised learning in marketing analytics. By grouping customers based on shared behavioral and demographic characteristics, businesses can design targeted campaigns, optimize resource allocation, and improve customer retention.

The objective of this project was to:

- Analyze and prepare a cleaned customer personality dataset for clustering
- Select relevant behavioral and demographic features for segmentation
- Determine the statistically optimal number of customer clusters
- Build, evaluate, and compare multiple clustering algorithms
- Profile each resulting segment and assign business-relevant names
- Validate cluster stability across different samples and random states
- Select the best-performing model and export a production-ready pipeline

2. Dataset Overview

Dataset: Customer Personality Final Dataset (pre-cleaned, standardized)

Records: 2,038 customers

Features: 40 columns (demographic, purchasing, and engagement variables)

Data Quality: 0 duplicate records and 0 missing values confirmed at the source dataset level; all features already standardized (mean ≈ 0, std ≈ 1)

Key feature groups present in the dataset include:

- Demographics: Education, Marital Status, Income, Age, Family Size, Total Children

- Purchase behavior: Amount spent on wines, fruits, meat, fish, sweets, and gold products
- Purchase channels: Web, catalog, and in-store purchase counts
- Engagement: Web visits, deal dependency, digital engagement, campaign response history
- Derived features: Total Spending, Total Purchases, Customer Tenure, Average Spending per Purchase

3. Methodology

3.1 Data Review & Cleaning

The dataset was reviewed for structural integrity prior to modeling. No duplicate rows or missing values were found in the full 40-column dataset. Scaling had already been applied upstream in the feature engineering stage, confirmed by descriptive statistics showing near-zero means and unit variances across numeric features.

3.2 Feature Selection for Clustering

A focused subset of 12 features was selected to drive the clustering process, chosen for their direct relevance to purchasing behavior and customer value:

- Customer_Age, Income, Recency, Customer_Tenure
- Family_size, Total_Children
- Total_spending, Total_Purchase, Total_Campaign
- NumWebPurchases, NumStorePurchases, NumCatalogPurchases

After feature selection, 2 duplicate records were identified within this reduced feature space and removed, leaving a clean modeling dataset.

3.3 Feature Scaling

As the dataset was already standardized in an earlier preprocessing stage, no additional scaling was required for the clustering-ready feature set.

3.4 Exploratory Cluster Analysis

Exploratory analysis was performed to understand feature relationships and natural groupings prior to model building, including:

- Pair plot (scatter matrix) of selected features
- Correlation heatmap to identify multicollinearity
- PCA-based 2D visualization to assess separability
- Feature distribution plots and boxplots to detect skewness and outliers

4. Determining the Optimal Number of Clusters

Three complementary statistical methods were used to determine the optimal cluster count (K), evaluated across K = 2 to 10:

4.1 Elbow Method (WCSS)

K	WCSS
2	46,562.31
3	41,448.74
4	39,198.68
5	37,515.45
6	36,430.46
7	35,526.93
8	34,822.41
9	34,069.14
10	33,491.77

4.2 Silhouette Score Analysis

K	Silhouette Score
2	0.2516
3	0.1844
4	0.1363
5	0.1121
6	0.1007
7	0.0991
8	0.0904
9	0.0834
10	0.0812

Best K (Silhouette): 2, with a peak score of 0.2516

4.3 Davies-Bouldin Index

K	Davies-Bouldin Index
2	1.5371
3	1.9120
4	2.2639
5	2.4172
6	2.5425
7	2.5034
8	2.5451
9	2.5638
10	2.5619

Best K (Davies-Bouldin): 2, with the lowest index value of 1.5371

All three evaluation methods consistently converge on K = 2 as the statistically optimal number of clusters, providing strong confidence in this choice for the final segmentation model.

5. Clustering Model Development & Evaluation

5.1 K-Means Clustering (Baseline Model)

Cluster Sizes: Cluster 0 — 1,111 customers | Cluster 1 — 927 customers

Silhouette Score: 0.2516

Davies-Bouldin Index: 1.5371

Cluster centers were computed across all standardized features, and results were visualized using both PCA and t-SNE for 2D interpretation. The trained model and clustered dataset were saved for downstream use.

5.2 Hierarchical (Agglomerative) Clustering

Cluster Sizes: Cluster 0 — 1,134 customers | Cluster 1 — 904 customers

Silhouette Score: 0.2380

Davies-Bouldin Index: 1.5873

A dendrogram was generated to visually confirm the natural split into two clusters, supporting the K = 2 recommendation from the earlier evaluation stage.

5.3 Gaussian Mixture Model (GMM)

Cluster Sizes: Cluster 0 — 1,150 customers | Cluster 1 — 888 customers

Silhouette Score: 0.2449

Davies-Bouldin Index: 1.5611

AIC: -179,353.88

BIC: -169,682.33

GMM additionally produced soft cluster-membership probabilities per customer, offering probabilistic insight into segment confidence rather than hard assignment alone.

5.4 DBSCAN Clustering

Result: 0 valid clusters formed; all 2,038 points classified as noise

With the parameters tested, DBSCAN was unable to identify meaningful density-based clusters in this feature space, and no silhouette score could be calculated. This indicates the customer data does not exhibit the density-based separation structure DBSCAN is designed to detect, and the algorithm was excluded from final model consideration.

6. Algorithm Comparison

Algorithm	Silhouette Score	Davies-Bouldin Index	Clusters	Interpretability	Stability
K-Means	0.2380	1.5873	2	High	High
Hierarchical	0.2380	1.5873	2	Medium	High
Gaussian Mixture	0.2449	1.5611	2	Medium	High
DBSCAN	N/A	N/A	0	Low	Low

Gaussian Mixture Modeling produced marginally stronger evaluation metrics than K-Means and Hierarchical Clustering, while DBSCAN failed to form usable clusters on this dataset. K-Means and Hierarchical Clustering produced near-identical results, reinforcing the robustness of the K = 2 segmentation.

7. Cluster Profiling & Business Segmentation

The final K-Means clusters were profiled against key business-relevant features (Income, Total Spending, Recency, Tenure, Family Size, and purchase channels) to translate statistical clusters into actionable customer personas.

Segment	Cluster Size	Relative Income	Relative Spending
High-Value Customers	1,111	Above average	Above average
Premium Buyers	927	Below average	Below average

Cluster 0 was labeled "High-Value Customers" — characterized by higher income and significantly higher total spending, representing the business's most valuable segment. Cluster 1 was labeled "Premium Buyers" — customers with comparatively lower income and spending levels, but still representing a substantial and addressable customer base. The final segmented dataset, including business-friendly segment labels, was exported for use by marketing and CRM teams.

8. Cluster Stability Analysis

To confirm the reliability of the K-Means segmentation, three independent stability tests were performed:

8.1 Different Random States

The model was re-trained with five different random seeds (0, 10, 20, 42, 100). The Silhouette Score remained constant at 0.2516 across all runs, confirming the clustering solution is not sensitive to initialization.

8.2 Different Training Samples (80% Subsampling)

Sample Seed	Silhouette Score
1	0.2524
10	0.2521
20	0.2506
42	0.2534
100	0.2533

8.3 Bootstrap Resampling

Bootstrap Sample	Silhouette Score
1	0.2496
2	0.2465
3	0.2500
4	0.2542
5	0.2554

All three stability tests produced tightly clustered Silhouette Scores (approximately 0.25 ± 0.005) with no significant variance, providing strong evidence that the K-Means segmentation is stable and reproducible across different samples and initializations.

9. Final Model Selection

Selected Algorithm: K-Means Clustering

Optimal Clusters: 2

Final Silhouette Score: 0.2380

Final Davies-Bouldin Index: 1.5873

Business Justification

K-Means was selected as the final production model for the following reasons:

- Achieved a strong, statistically validated Silhouette Score
- Achieved a low Davies-Bouldin Index, indicating well-separated clusters
- Produced clearly distinguishable, business-interpretable customer segments
- Simple to understand, implement, and explain to non-technical stakeholders
- Computationally efficient and scalable for production deployment
- Demonstrated the highest stability across random-state, sampling, and bootstrap tests

10. Production Pipeline & Deployment Artifacts

The following artifacts were generated and saved to support production deployment of the segmentation model:

Artifact	Status
Feature list (feature_list.pkl)	Saved
Standard scaler (standard_scaler.pkl)	Saved
Preprocessing pipeline (preprocessing_pipeline.pkl)	Pending
Final clustering model (final_clustering_model.pkl)	Pending
K-Means model (kmeans_model.pkl)	Saved
Final segmented customer dataset (CSV)	Saved
Algorithm comparison table (CSV)	Saved
Cluster stability reports (CSV)	Saved
Final model recommendation summary (CSV)	Saved

Two production artifacts — the consolidated preprocessing pipeline and the final packaged clustering model file — remain outstanding and are recommended as immediate next steps before full deployment.

11. Conclusion & Recommendations

This project successfully delivered a complete, validated customer segmentation solution. Through systematic evaluation of four clustering algorithms and rigorous statistical validation, two stable and business-meaningful customer segments were identified: High-Value Customers and Premium Buyers.

Recommendations for next steps:

- Finalize and save the consolidated preprocessing pipeline and packaged production model
- Integrate the segmentation output into CRM and marketing automation systems
- Design segment-specific marketing campaigns, prioritizing retention strategies for High-Value Customers
- Re-run the clustering pipeline periodically (e.g., quarterly) to capture shifts in customer behavior
- Investigate alternative distance metrics or parameter tuning for DBSCAN if density-based segmentation is required in future work